

2. REQUIREMENT ANALYSIS PHASE

ComicCraft Project Documentation & Specification

Document: Requirements Analysis & System Design

Project: ComicCraft AI

1. Introduction

Requirement analysis converts the foundational concept into clear functional and technical requirements. ComicCraft must collect user information, process it through the AI workflow, display the generated comic dynamically, and export the final output as a downloadable PDF.

2. Objectives

- Identify and structure user inputs for optimal prompt engineering.
- Define core functional requirements for narrative and visual generation.
- Define non-functional requirements such as security, execution efficiency, and UI responsiveness.
- Identify software and package dependencies required for backend and frontend workflows.
- Define API endpoint architectures and dynamic frontend rendering requirements.
- Define expected outputs, error handling, and data validation standards.

3. Phase Workflow

Inputs

- Story Prompt
- Character Name
- Setting & Background
- Tone & Mood
- Art Style Selection

Outputs

- Panel-by-Panel Story Script
- AI Generated Illustrations
- Interactive Comic Preview
- Downloadable PDF Document

4. Detailed Explanation

The technical architecture relies on **FastAPI** and **Uvicorn** for high-performance RESTful services in Python. The UI layer is implemented using **HTML/CSS/Jinja2** templates. Core computational engines utilize **Google Generative AI** for plot generation, alongside **Hugging Face Diffusers**, **PyTorch**, and **FPDF** for image generation and PDF compilation.

Security Guideline: API keys and sensitive credentials must never be committed directly to GitHub. Store sensitive variables in an uncommitted `.env` file and distribute a sanitized `.env.example`.

5. GitHub-Ready Code / Structure

Below is the data validation schema for incoming prompt requests:

```
from pydantic import BaseModel

class PromptRequest(BaseModel):
    story_prompt: str
    character_name: str
    setting: str
    tone: str
    art_style: str

def validate_prompt(data: PromptRequest):
    if not data.story_prompt.strip():
        raise ValueError("Story prompt is required")
    return data
```

6. GitHub Files / Deliverables

File Path	Purpose / Description
docs/requirements.md	Comprehensive documentation of system requirements and scope.
app/schemas.py	Pydantic data validation models and schema specifications.
.env.example	Template for environment variables and configuration secrets.
requirements.txt	Dependency list including FastAPI, Diffusers, PyTorch, and FPDF.

7. Expected Outcome

At the end of Phase 2, the corresponding part of ComicCraft should be clearly documented, implemented, or verified according to the purpose of the phase. The actual completion status should be based on the student's repository setup and verified execution flow.

8. Phase Conclusion

Phase 2 provides a structured foundation for the next stage of ComicCraft development, keeping the project organized, reproducible, and ready for GitHub-based collaboration and academic evaluation.